

1. (a) Since
- $\cos x = 2 \cos^2 \frac{1}{2}x - 1$

$$\int_0^\pi \sqrt{1 + \cos x} dx = \int_0^\pi \sqrt{1 + 2 \cos^2 \frac{1}{2}x - 1} dx = \int_0^\pi \sqrt{2 \cos^2 \frac{1}{2}x} dx = \sqrt{2} \int_0^\pi \cos \frac{1}{2}x dx$$

$$\text{Let } u = \frac{1}{2}x \Rightarrow du = \frac{1}{2}dx$$

$$\text{and } x = 0 \Rightarrow u = 0 ; x = \pi \Rightarrow u = \frac{1}{2}\pi$$

$$\text{So } \int_0^\pi \sqrt{1 + \cos x} dx = \sqrt{2} \int_0^\pi \cos \frac{1}{2}x dx = \sqrt{2} \int_0^{\frac{\pi}{2}} 2 \cos u du = 2\sqrt{2} [\sin u]_0^{\frac{\pi}{2}} = 2\sqrt{2}$$

- (b) Let
- $u = 1/x \Rightarrow du = dx/x^2$

$$\text{and } x = 1 \Rightarrow u = 1 ; x = 2 \Rightarrow u = 1/2$$

$$\int_1^2 \frac{e^{1/x}}{x^2} dx = \int_1^{1/2} -e^u du = [-e^u]_1^{1/2} = e - \sqrt{e}$$

$$(c) \int_0^{\frac{\pi}{4}} \frac{\arctan x}{1+x^2} dx = \frac{1}{2} [(\arctan x)^2]_0^{\frac{\pi}{4}} = \frac{1}{2}$$

$$(d) \int \frac{dx}{x^2 - 4x + 13} = \int \frac{dx}{(x-2)^2 + 9} = \frac{1}{3} \arctan\left(\frac{x-2}{3}\right) + C$$

- (e) Let
- $u = \ln x \Rightarrow du = dx/x$

$$\int \frac{1}{x \ln x} dx = \int \frac{1}{u} du = \ln |u| + C = \ln |\ln x| + C$$

$$(f) \int_{-1}^1 \frac{x^2}{1+x^2} dx = \int_{-1}^1 \frac{1+x^2-1}{1+x^2} dx = \int_{-1}^1 \left(1 - \frac{1}{1+x^2}\right) dx = [x - \arctan x]_{-1}^1 = 2 - \frac{\pi}{2}$$

$$(g) \int_0^{\frac{1}{2}} \frac{1+x}{\sqrt{1-x^2}} dx = \int \frac{x}{\sqrt{1-x^2}} dx + \int \frac{dx}{\sqrt{1-x^2}} = [-\sqrt{1-x^2} + \arcsin x]_0^{\frac{1}{2}} = 1 - \frac{\sqrt{3}}{2} + \frac{\pi}{6}$$

$$(h) \text{ Note that } \frac{x}{1-x^4} = \frac{x}{(1+x^2)(1+x)(1-x)} = \frac{Ax+B}{1+x^2} + \frac{C}{1+x} + \frac{D}{1-x}$$

$$x = (Ax+B)(1-x^2) + C(1+x^2)(1-x) + D(1+x^2)(1+x)$$

$$\text{when } x = 1, \text{ then } 1 = 4D \Rightarrow D = \frac{1}{4}$$

$$\text{when } x = -1, \text{ then } -1 = 4C \Rightarrow C = -\frac{1}{4}$$

$$\text{when } x = 0, \text{ then } 0 = B + C + D \Rightarrow B = 0$$

$$\text{when } x = 2, \text{ then } 2 = -6A - 5C + 15D \Rightarrow A = \frac{1}{2}$$

$$\int \frac{x}{1-x^4} dx = \frac{1}{2} \int \frac{x}{1+x^2} dx - \frac{1}{4} \int \frac{1}{1+x} dx + \frac{1}{4} \int \frac{1}{1-x} dx$$

$$= \frac{1}{4} (\ln |1+x^2| - \ln |1+x| + \ln |1-x|) + C$$

- (i) Let
- $u = \ln x \Rightarrow du = dx/x$

$$\text{and } x = 1 \Rightarrow u = 0 ; x = c \Rightarrow u = \ln c$$

$$\int_1^c \frac{\ln x^3}{x} dx = \int_1^c \frac{3 \ln x}{x} dx = \int_0^{\ln c} u du = 3 \left[ \frac{u^2}{2} \right]_0^{\ln c} = \frac{3(\ln c)^2}{2}$$

$$(j) \int a^x e^x dx = \int (ae)^x dx = \frac{(ae)^x}{\ln(ae)} + C = \frac{a^x e^x}{1 + \ln a} + C$$

- (k) Let
- $u = x \Rightarrow du = dx ; dv = 2^x dx \Rightarrow v = 2^x / \ln 2$

$$\int_0^2 x 2^x dx = \frac{x 2^x}{\ln 2} - \frac{1}{\ln 2} \int_0^2 2^x dx = \frac{1}{\ln 2} [x 2^x - \frac{2^x}{\ln 2}]_0^2 = \frac{1}{\ln 2} (8 - \frac{3}{\ln 2})$$

(l) Let  $u = x^2 \Rightarrow du = 2x dx$  ;  $dv = (2x - 1)^{-7} dx \Rightarrow v = \frac{(2x-1)^{-6}}{-12}$

$$\begin{aligned} \int x^2 (2x - 1)^{-7} dx &= -\frac{x^2}{12} (2x - 1)^{-6} + \int \frac{x}{6} (2x - 1)^{-6} dx \\ &= -\frac{x^2}{12} (2x - 1)^{-6} - \frac{x}{60} (2x - 1)^{-5} + \int \frac{1}{60} (2x - 1)^{-5} dx \\ &= -\frac{x^2}{12} (2x - 1)^{-6} - \frac{x}{60} (2x - 1)^{-5} - \frac{1}{480} (2x - 1)^{-4} + C \end{aligned}$$

(m) Let  $u = \ln(1 + x^2) \Rightarrow du = \frac{2x}{1 + x^2} dx$  ;  $dv = dx \Rightarrow v = x$

$$\begin{aligned} \int \ln(1 + x^2) dx &= x \ln(1 + x^2) - 2 \int \frac{x^2}{1 + x^2} dx = x \ln(1 + x^2) - 2 \int \frac{x^2 + 1 - 1}{1 + x^2} dx \\ &= x \ln(1 + x^2) - 2 \int (1 - \frac{1}{1 + x^2}) dx = x \ln(1 + x^2) - 2x + 2 \arctan x + C \\ \int_0^1 \ln(1 + x^2) dx &= [x \ln(1 + x^2) - 2x + 2 \arctan x]_0^1 = \ln 2 - 2 + \frac{\pi}{2} \end{aligned}$$

(n) Let  $u = \ln x \Rightarrow du = \frac{1}{x} dx$  ;  $dv = x^n dx \Rightarrow v = \frac{1}{n+1} x^{n+1}$

$$\int x^n \ln x dx = \frac{1}{n+1} x^{n+1} \ln x - \int \frac{1}{n+1} x^n dx = \frac{1}{n+1} x^{n+1} \ln x - \frac{x^{n+1}}{(n+1)^2} + C$$

(o) By using integration by parts formula 3 times, we can get

$$\begin{aligned} \int x^3 \sin x dx &= -x^3 \cos x + 3 \int x^2 \cos x dx = -x^3 \cos x + 3x^2 \sin x - 6 \int x \sin x dx = \\ &= -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C \end{aligned}$$

(p) Let  $u = \sqrt{x} \Rightarrow du = \frac{1}{2\sqrt{x}} dx$  ;  $dv = \frac{\cos \sqrt{x}}{\sqrt{x}} dx \Rightarrow v = 2 \sin \sqrt{x}$

$$\int \cos \sqrt{x} dx = \int \sqrt{x} \frac{\cos \sqrt{x}}{\sqrt{x}} dx = 2\sqrt{x} \sin \sqrt{x} - \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = 2\sqrt{x} \sin \sqrt{x} + 2 \cos \sqrt{x} + C$$

(q) Let  $u = \ln x \Rightarrow du = \frac{1}{x} dx$

$$\begin{aligned} \int \frac{1}{x} \arcsin(\ln x) dx &= \int \arcsin u du = u \arcsin u + \sqrt{1 - u^2} + C \\ &= (\ln x) \arcsin(\ln x) + \sqrt{1 - (\ln x)^2} + C \end{aligned}$$

(r) Let  $u = \cos(\ln x) \Rightarrow du = -\frac{\sin(\ln x)}{x} dx$  ;  $dv = dx \Rightarrow v = x$

$$\int \cos(\ln x) dx = x \cos(\ln x) + \int \sin(\ln x) dx$$

Let  $u = \sin(\ln x) \Rightarrow du = \frac{\cos(\ln x)}{x} dx$  ;  $dv = dx \Rightarrow v = x$

$$\int \cos(\ln x) dx = x \cos(\ln x) + \int \sin(\ln x) dx = x \cos(\ln x) + x \sin(\ln x) - \int \cos(\ln x) dx$$

$$\Rightarrow 2 \int \cos(\ln x) dx = x \cos(\ln x) + x \sin(\ln x)$$

$$\Rightarrow \int \cos(\ln x) dx = \frac{1}{2} x [\cos(\ln x) + \sin(\ln x)] + C$$